

Jeremy Liu

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Education

University of Toronto

Bachelor of Applied Science in Computer Engineering + PEY Co-op

Sept 2025 – May 2030

Toronto, ON

Technical Skills

Languages: Python, JavaScript, TypeScript, C, MATLAB, HTML, CSS

Frameworks: React, Next.js, Express.js, TailwindCSS, Bootstrap

Libraries: Pandas, NumPy, OpenCV, Selenium, BeautifulSoup, Matplotlib

Developer Tools: Node.js, AWS, PostgreSQL, Git, Arduino, Visual Studio Code, Cursor.ai

Projects

Stop! Don't Go On! (Finalist, GoOnHacks 2025) | *Arduino, Python, Flask, React, OpenCV*

Nov 2025

- Built a computer vision application to detect prolonged inactivity and attention loss to boost productivity for over 40+ users.
- Implemented real-time face tracking using OpenCV and triggered immediate hardware-based alerts using Arduino.
- Designed and served REST APIs using Flask to coordinate frontend and backend communication.
- Created a dynamic UI that allowed user input and real-time computer feedback using React, Vite, and Typescript.

Binder. | *Python, Selenium, BeautifulSoup, React, TailwindCSS, Gemini API*

Oct 2025

- Built a swipe-based website that browses second-hand marketplaces in a more intuitive and engaging way.
- Used BeautifulSoup to scrape for data collection in real-time and leveraged the Gemini AI API for price evaluation.
- Developed price analysis and AI negotiation guidance features based on historical listing data.
- Built responsive UI components using React, TypeScript and TailwindCSS.

UFC Index | *Python, JavaScript, Pandas, BeautifulSoup, Next.js, React, TailwindCSS, TypeScript*

Dec 2025

- Used Python and BeautifulSoup to parse ufcstats.com, while using safe rate limiting and error handling practices.
- Implemented an elo rating algorithm in Python with Pandas to generate statistically accurate performance metrics and ranked leaderboards for 2500+ fighters.
- Used RESTful API routes to serve CSV data as JSON endpoints, creating real-time visualizations.
- Developed a production-ready Next.js application with an responsive UI using React, Typescript, and TailwindCSS.

j-space | *HTML, CSS, JavaScript, Vite, Component Libraries*

Dec 2025

- Used HTML, CSS, Javascript, and Vite to create a fully functional, interactive Windows 98 desktop environment.
- Integrated and customized open-source libraries to implement authentic Windows 98 UI components.
- Engineered responsive design patterns to ensure cross-browser compatibility and optimal user experience.

Experience

Software Engineer

Sep 2025 - Present

Robotics for Space Exploration

Markham, ON

- Developed robotics software in Python to optimize robot performance for student competitions.
- Wrote unit tests for ROS-based robotics components with a Linux VM.
- Collaborated with cross-functional teams to deliver scalable, mission-driver projects under strict time constraints.
- Implemented a ROS2-PyQt GUI for a robotic arm, providing real-time joint state monitoring, camera feeds, emergency stop control, and interactive 3D-path-planning visualization.

Private Tutor

Sep 2023 – Present

Math, Computer Science, Physics, Chemistry, English

Toronto, ON

- Improved academic performance for 30+ students by over 20% through personalized lesson plans and targeted problem-solving strategies.
- Adapted explanations to diverse learning styles, strengthening student engagement and conceptual understanding.
- Managed scheduling, lesson planning, and iterative feedback to ensure consistent progress and measurable outcomes.